CTRL+EXP+DEL:
The Domestic Costs of U.S. Export Controls

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Abstract

National-security restrictions on exports constrain exporters' market access, but empirical evidence on their domestic costs is limited. We leverage data on export transactions that includes the product classifications used to administer U.S. export controls to document a doubling in the share of U.S. exports subject to controls between 2010 and 2024 and a skewed firm-size distribution among firms that export controlled products. We then exploit the 2014 U.S. restriction on exports to Russia as a natural experiment to examine the impact of export controls on firm-level outcomes. These controls not only reduced U.S. firms' probability of exporting targeted goods to Russia but also had indirect effects: they reduced affected firms' exports of other products and to other destinations and lowered affected firms' average payroll. Our findings imply that the domestic costs of export controls extend beyond direct restrictions.

Keyword: Sanctions, Export Controls, U.S.-Russia Trade

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1 Introduction

Governments increasingly rely on export controls to restrict sales of products with both commercial and military applications, often called "dual-use" products (Clayton, Maggiori, and Schreger, 2025). However, the domestic costs of export controls remain largely unexplored. The empirical challenge is that export controls regulate products based on technical specifications and intended use rather than the physical characteristics captured by standard trade classifications. Without data distinguishing controlled from uncontrolled products, researchers cannot assess how controls affect trade patterns, which firms bear the costs of complying and documenting compliance with the regulations, or how firms adjust behavior when controls are introduced or expanded. We overcome this challenge using newly available public data on export transactions containing the product codes U.S. regulators use to administer controls and confidential customs transactions by firms.

Our analysis proceeds in three steps to evaluate the domestic impact of U.S. export controls. We document the prevalence and evolution of controlled exports to establish the scope of the policy; we characterize the firms that export controlled products to understand exposure patterns; and we provide firm-level estimates of direct and indirect costs. We establish three main results.

First, the share of U.S. export value that is subject to export controls doubled from about 30% to 60% between 2010 and 2024. This increase is consistent with a broadening of coverage of exports controls, rather than an increase in exports of specific controlled products. Most of that increase was driven by exports subject to regulation but which did not require export licenses and by exports to destinations that receive the most favorable treatment under the U.S. export-control system.

Second, leveraging confidential customs data from the U.S. Census Bureau, we examine the characteristics of firms that export controlled products. We use firm size as an organizing principle to understand patterns of participation in dual-use markets and infer the nature of compliance costs. Like overall exports, controlled exports are disproportionately concentrated in a small fraction of firms. This similarity implies that the cost of compliance with export controls do not fundamentally reshape the firm-size distribution in dual-use export markets. Among firms exporting controlled products, small firms tend to export items requiring less extensive regulatory review, whereas large firms maintain more diversified product portfolios. Because firm size correlates strongly with multinational status and R&D expenditures, we also explore these margins of firm heterogeneity. We document that related-party (i.e., within-firm) transactions are more common for controlled exports than for other exports and that R&D-performing firms account for a disproportionate share

of controlled exports. These findings are consistent with the extensive firm-heterogeneity literature documenting that the largest and most productive firms dominate export markets (Bernard, Jensen, Redding, and Schott, 2007; Bernard, Jensen, and Schott, 2009), that fixed costs create size-based advantages (Melitz, 2003; Roberts and Tybout, 1997), and that firm size, multinational status, and R&D activity are jointly determined characteristics of high-productivity exporters (Antràs and Yeaple, 2009; Aw, Roberts, and Xu, 2011).

Finally, using confidential firm-level customs transactions, we provide the first empirical evidence of the impact of U.S. export controls on domestic firms' economic activities through a case study of the U.S. sanctions on exports to Russia following Russia's February 2014 invasion and occupation of Crimea. Specifically, we examine the impact of Russia-targeted export controls that became effective August 6, 2014 (U.S. Department of Commerce, Bureau of Industry and Security, August 6, 2014, hereafter FR 45675). These Russia-targeted controls prohibited exports of products that may be used directly or indirectly for the exploration or production of oil from deep-water, Arctic offshore, or shale projects in Russia. Using a difference-in-differences approach, we compare firms that previously exported targeted goods to Russia with those that exported the same products to China, to isolate the causal impact of the Russia-targeted export controls on U.S. exporters. We find that the sanctions had both direct effects—reducing impacted firms' probability of exporting targeted goods to Russia—and indirect effects—reducing their value of exports of other goods, value of exports to other destinations, and payroll. These results are consistent with complementarities in exports across related products and to other destinations and rule out simple mechanical explanations like large-scale product-code reclassification or strategic diversion of controlled exports to other destinations.

Our analysis is most closely related to two recent papers examining the domestic impacts of U.S. export controls, although different in approach and scope. Fan, Handley, and Limão (2025) calibrate the economy-wide welfare costs of U.S. export controls using aggregate controlled export patterns in a quantitative trade model. Our firm-level perspective complements their approach by revealing which firms bear the compliance costs and how these firms adjust exports across products and destinations. Understanding firm-level responses to export controls enables an assessment of their distributional impacts. Crosignani, Han, Macchiavelli, and Silva (2026) examine export controls targeting Chinese entities using 176 U.S. supplier–Chinese buyer pairs and find that these controls cause U.S. firms to stop trading

with both targeted and non-targeted customers and reduce profitability and employment.[1] Our case study of U.S.-imposed controls on exports to Russia finds similar patterns and in a setting in which we can identify the complete set of impacted U.S. firms and construct a clean control group of exporters selling identical products to an unaffected destination. In addition, observing the near-universe of U.S. exporters enables us to establish stylized facts—large-firm dominance, small-firm specialization in controlled products—infeasible to detect in small samples.

Our focus on export controls complements research that examines the impacts of broader trade sanctions on sanctioning countries.[2] This literature documents costs on non-U.S. Western firms that vary systematically with firm characteristics and generate unintended spillovers that extend beyond directly targeted products.[3] For example, Crozet and Hinz (2020) document aggregate trade losses for Western firms that were subject to sanctions due to their trade with Russia in 2014 and find evidence of export losses for sanctioning countries' exporters. In a case study of French firms, the authors find that a loss of exports of non-embargoed products to Russia was associated with a decline in the provision of trade finance. Crozet, Hinz, Stammann, and Wanner (2021) focus on French firms' extensive-margin export decisions across four sanctions episodes (on exports to Iran, Russia, Myanmar, and Cuba) and also find heterogeneous trade-sanctions effects: larger firms and those that rely heavily on trade-finance instruments face greater barriers to market entry under sanctions, whereas firms with experience exporting to politically unstable markets are more resilient. Gullstrand (2020) finds that smaller Swedish firms exit sanctioned markets but that larger multinationals maintain a presence in these markets despite experiencing losses. We extend this literature by examining the impacts of U.S. export controls—a distinct but increasingly important policy tool—on U.S. exporters.

The rest of the paper is organized as follows. Section 2 provides institutional background on U.S. export controls and documents their increased use over the past 15 years. Section 3 describes the Census Bureau microdata. Section 4 provides summary statistics on firms that export controlled products. The case study of the impact of the Russia-targeted export controls following the 2014 invasion of Crimea is presented in Section 5. Section 6

[1] A parallel literature examines the effects of various waves of U.S. export controls on targeted Chinese firms (e.g., Albertson, Howard, and Sarzosa, 2026; Liu, Liu, Makarin, and Wen, 2025; Liu, Liu, and Zhang, 2026).

[2] Examples of studies examining the impact of financial sanctions on sender-country firms include Besedeš, Goldbach, and Nitsch (2017, 2021), who document these effects using firm-level data from Germany.

[3] Related research documents broad impacts of Western sanctions on Russia in 2014 (e.g., Yalcin, Felbermayr, Kariem, Kirilakha, Kwon, Syropoulos, and Yotov, 2025; Ahn and Ludema, 2020; Nigmatulina, 2025; Huynh, Hoang, and Ongena, 2025) and in 2022 (e.g., Egorov, Korovkin, Makarin, and Nigmatulina, 2025).

concludes with a discussion of potential channels through which export controls impose costs on domestic firms.

2 U.S. Export Controls

The U.S. export-control system consists of several licensing and enforcement agencies, including the Department of Commerce, which regulates and licenses exports of dual-use goods and technologies; the Department of State, which regulates arm exports; and the Department of the Treasury, which administers financial sanctions (Congressional Research Service, 2021, p. 1). This paper focuses on the exports of products that fall under the jurisdiction of the Department of Commerce.

Export Administration Regulations

The Bureau of Industry and Security (BIS), an agency within the Department of Commerce, administers the export-licensing and enforcement functions of the dual-use export-control system to advance U.S. national-security, foreign-policy, and economic objectives.[4] The BIS enforces and implements the Export Administration Regulations (EAR), which regulate the export of dual-use goods and technology. These include less-sensitive military products, commercial products that have both commercial and military applications, and some commercial products that do not have any obvious military use (U.S. Department of Commerce, Bureau of Industry and Security, 2025d).

U.S. exporters have the primary responsibility to ensure that their products comply with the EAR. An export of a product is subject to controls under EAR if it is on the Commerce Control List (CCL), a catalog maintained by BIS of products subject to EAR. Each product in the CCL is associated with an Export Control Classification Number (ECCN), a five-character alphanumeric code that identifies the specific product and reasons for its control.[5] Reasons for control include concerns about national security, chemical and biological weapons, nuclear proliferation, and regional stability (U.S. Department of Commerce, Bureau of Industry and Security, 2025b). If a product is subject to the EAR but does not appear on the CCL, its ECCN is designated "EAR99."

[4] All acronyms used in the text are summarized in Table 1.

[5] See Appendix A for more details on ECCNs. ECCN data are unavailable for U.S. exports to Canada because these statistics are based on Canadian import records under a 1987 bilateral agreement (Mozes and Oberg, 2002); we therefore exclude exports to Canada throughout our analysis.

For products on the CCL, U.S. exporters may have to obtain an export license or a license exception from BIS. EAR99-designated shipments generally do not require licenses, but a license may be required if the product is exported to a prohibited end user, for a restricted end use, or to a sanctioned or embargoed destination. The implication is that U.S. exporters must perform careful due diligence to ensure compliance with export regulations by classifying their products accurately and determining whether a license is required.

Licensing Requirements

License requirements depend not only on the technical characteristics of a product delineated by the ECCN but also its destination, end user, and end use. Therefore, even a product not listed in the CCL and designated as EAR99 (e.g., a low-technology consumer good) may require a license if it is being exported to an embargoed country, to an end user of concern, or in support of a prohibited end use. The Commerce Country Chart contains information on reasons for export control by destination (U.S. Department of Commerce, Bureau of Industry and Security, 2025b). Using the reasons for export control described by the ECCN, exporters can look up the analogous reason for control in the chart and determine whether a license is required for exports to a particular destination. In addition to the country control chart, exporters must also consult four separate lists maintained by the BIS (Denied Persons List, Entity List, Unverified List, and Military End-User List) that classify end users with specific licensing requirements (U.S. Department of Commerce, Bureau of Industry and Security, 2025f).[6]

License exceptions—authorizations enabling export of products subject to the EAR that would otherwise require a license—may be available based on country-group classifications (U.S. Department of Commerce, Bureau of Industry and Security, 2025c). Countries are classified into four groups: A, B, D, and E, with broadly increasing stringency of controls (see U.S. Department of Commerce, Bureau of Industry and Security, 2025j, and Appendix B for additional details). Most countries are included in Group B. These are countries with which the United States has a basic level of trust with respect to trade and are subject to relatively lenient license-review policies. Countries in Group A form a subset of Group B countries; they are considered to pose a low risk of unauthorized export or diversion of controlled items and receive the most favorable treatment available in the EAR. Group D countries (some of which also appear in Group B) consist of those countries identified with

[6] A consolidation of lists of individuals and organizations subject to trade-related restrictions, known as the Consolidated Screening List, is maintained by the Departments of Commerce, State, and the Treasury (U.S. Department of Commerce, 2025).

specific proliferation concerns and subject to U.S. arms embargoes that are administered by the Department of State. Group E countries are designated as state sponsors of terrorism and are subject to a unilateral embargo. Not all countries have a group designation; some three dozen island nations, including Bermuda, Faroe Islands, and Niue, do not have a country group.[7] For countries assigned to more than one group, we use the most restrictive licensing requirements. For example, Taiwan and Israel are listed in groups A, B and D, so we include them in Group D only. Because all countries in Group A are also in Group B, this reduces the number country groups to three: B, D, and E.

Export licenses issued by the BIS govern the terms under which products subject to the EAR may be exported. Standard licenses are valid for four years from the date of issuance and specify both a maximum dollar value and quantity of goods that may be exported.[8] Critically, each license is tied to a single end user and end use, effectively binding the exporter to a specific buyer and application. Although firms may request extensions for multi-year projects or when production lead times exceed the standard four-year window, routine transactions—including continuing requirements for spare parts—do not qualify for extensions. BIS regulations permit combining multiple items on a single license application, but applications covering dissimilar products may face lengthier review processes as they are referred to multiple BIS offices (U.S. Department of Commerce, Bureau of Industry and Security, 2026a). This feature of the licensing regime creates incentives for firms to separate applications by product category, potentially increasing administrative burdens for diversified exporters. The specificity of license terms means that any change in buyer, destination, or intended use requires a new application or formal amendment, adding ongoing compliance costs.

Trends in U.S. Controlled Exports, 2010–2024

Figure 1(a) shows the share of total U.S. export value, quarterly from 2010 through 2024, subject to the EAR. [9] The share of U.S. export value with ECCNs is on average 13% of all export value. It reached a high of around 15% in 2016 following the transition, beginning

[7] In the following sections, we include countries without a group designation when we present statistics on all exports, but omit them from group breakdowns.

[8] See U.S. Department of Commerce, Bureau of Industry and Security (2026b) for licensing procedures; see U.S. Department of Commerce, Bureau of Industry and Security (2026a) for complete administrative requirements for export applications.

[9] The data from U.S. Census Bureau (2024) start in 2009, but the first year exhibits patterns inconsistent with subsequent years: more than 200 ECCNs appear only in 2009. To ensure comparability over time, all analyses using U.S. Census Bureau (2024) start in 2010.

in 2013, of certain items that were previously controlled by the Department of State under the United States Munitions List to the CCL (U.S. Department of Commerce, Bureau of Industry and Security, 2026c).[10] In contrast, the share of export value designated as EAR99, increased from 23% of all export value to about 45% of all export value over this period. Combined, the value of exports subject to export controls increased from about 33% in 2010 to about 60% in 2024.

Figures 1(b), (c), and (d) repeat this exercise for the three primary country-group designations. The pattern in worldwide exports is largely driven by exports to Group B countries, but across all three groups, ECCN exports account for 10–20% of all export value, and EAR99 designations have increased dramatically over the period studied.

Group B countries—comprising U.S. allies and partner nations—account for the vast majority of exports subject to the EAR, with their share of export value increasing slightly from 76.3% to 77.3% between 2010 and 2024. Group D countries, which include Russia and China among other nations subject to heightened restrictions, correspondingly saw their share decline modestly from 21% to 18.9% over this period. Group E countries under comprehensive embargo (Cuba, North Korea, Syria, and Iran) represent a negligible share of exports subject to the EAR—0.25% in 2010 and 0.06% in 2024—reflecting the near-total prohibition on trade with these destinations. The similarity of the time series across the three country groups suggests that the doubling of controlled-export shares reflects a broadening of product coverage, rather than a reorientation of exports subject to the EAR toward or away from specific countries or country groups. We discuss the patterns specific to Russia and China in Appendix C, two countries in Group D, which serve as our primary treated and control destinations, respectively, in the firm-level case study in Section 5.

The aggregate trends establish that export controls cover over half of U.S. export value by 2024, almost double their 2010 share. However, aggregate statistics cannot reveal which firms bear compliance costs or how firms respond when controls tighten. In the next section, we describe the confidential Census Bureau microdata linking firms' export transactions to domestic operations that enable a firm-level analysis.

3 Firm-Level Data

We use three confidential datasets maintained by the U.S. Census Bureau for our firm-level analysis. First, the Longitudinal Firm Trade Transactions Database (LFTTD) includes all

[10]The CCL added the 600-series ECCNs, 9x515 ECCNs for Spacecraft and Satellites, and ECCNs for firearms (U.S. Department of Commerce, Bureau of Industry and Security, 2017).

goods export shipments valued over US$2,500 beginning in 1992, augmented with firm identifiers (U.S. Census Bureau, 2026; Kamal and Ouyang, 2020).[11] For each export shipment, we observe the date, destination, ten-digit Schedule B code used to classify goods for exports from the United States, ECCN or EAR99 designation (if one is associated with the shipment), and whether the transaction takes place between related parties using an indicator for related-party trade.[12] ECCNs do not correspond to the standard Schedule B product codes.[13] Whereas the latter are based on the product's physical characteristics, ECCNs are determined based on the technical specifications and intended use of the product. The vast majority of ten-digit Schedule B codes, including many seemingly innocuous codes such as those indicating apparel products, are associated with at least some shipments of controlled products (see Appendix A for ECCN–Schedule B mappings). The LFTTD does not contain information on the foreign buyer, which precludes identifying export transactions to foreign legal persons on the Entity List. License information is sometimes, but not always, included in the LFTTD, so we cannot comprehensively identify firms with export licenses (see Appendix D for details). Thus, we identify firms exporting shipments subject to the EAR by their ECCN or EAR99 designation.

Second, we combine information on firm-level exports in the LFTTD with firm-level domestic payroll and employment in the 2023 Longitudinal Business Database (LBD; see Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White, 2021).[14] LBD employment is measured as of the week of March 12 of the calendar year and payroll is measured over the full year.

Finally, we also link the LFTTD to the 2023 Business Enterprise Research and Development (BERD) survey to identify firms' research-and-development (R&D) status. The survey sampled about 46,000 firms, selected using a stratified design intended to capture as many R&D performers as possible. Firms that are known to invest heavily in R&D and firms with high payroll are selected with certainty, and other firms are sampled with a probability that

[11] A shipment is an electronic export information filing that contains mandatory information on an export transaction filed through the Automated Export System (AES) (U.S. Customs and Border Protection, 2026). We use LFTTD version 202301 for the analysis in Section 4 and the 201901 version for the analysis in Section 5. The first four digits of the version number indicate the last year of data available.

[12] The indicator denotes relationships in which one firm owns a stake of at least 10% in the other as reported by U.S. exporters in their export filings.

[13] Schedule B codes, administered by the U.S. Census Bureau (U.S. Census Bureau, 2025c), are ten-digit codes where the first six-digits follow the Harmonized System (HS) codes (World Customs Organization, 2025).

[14] We use version 202301 of the LBD for the analysis in Section 4. We use version 202100 of the LBD for the analysis in Section 5. As with the LFTTD, the first four digits of the version number indicate the last year of data available.

depends on past R&D performance (if known) or payroll (National Center for Science and Engineering Statistics, 2025). We classify firms with nonzero, nonmissing, R&D values as known R&D performers; about 6,400 of these had one or more export shipments in 2023. We cannot confirm whether the remaining 185,000 exporters have any R&D expenditures; in addition to firms that perform no R&D, these include firms that are excluded from the BERD survey as well as firms that were selected but did not respond to the survey.

4 A Portrait of Firms Exporting Controlled Products

We examine firm heterogeneity along three dimensions that have been previously shown to correlate with export status: size, multinational status, and R&D status. In standard trade models, only highly productive firms overcome the fixed and variable costs of entering foreign markets, creating a hierarchy in which large firms export diverse products to many destinations and small firms serve limited markets (Roberts and Tybout, 1997; Melitz, 2003; Bernard, Jensen, Redding, and Schott, 2007). Export controls add regulatory complexity to these costs by requiring product-classification expertise, licensing procedures, documentation systems, and ongoing compliance monitoring. If these compliance costs are substantial and include fixed components, they may amplify existing participation patterns that favor large firms. We examine firm heterogeneity in controlled export patterns, using the 2023 LFTTD, LBD, and BERD. We rely on size as a proxy for firms' ability to absorb compliance costs (e.g., financial resources, legal infrastructure, and regulatory experience).

Concentration and the Distribution of Controlled Exports

We begin by assessing whether exports subject to the EAR are more concentrated among large firms than general exports. If compliance imposes substantial fixed costs, such as establishing internal compliance programs and training specialized staff, we would expect exports subject to the EAR to exhibit greater concentration among large firms than overall exports. Alternatively, similar or lower concentration patterns would suggest that compliance costs are either modest or scale proportionally with firm size. Table 2 shows that the top 1% of U.S. exporters account for 84% of all export value, with the top 10% capturing 97%. These patterns are similar to those in 2007, as documented by Bernard, Jensen, Redding, and Schott (2018). Exports subject to the EAR exhibit nearly identical concentration: the top 1% of firms exporting controlled products account for 88% of such exports, and the top 10% account for 98%. When we restrict attention to explicit ECCN exports, concentration remains comparable: 82% and 98%, respectively. This similarity in concentration suggests

that export-control compliance does not create implicit barriers to entry in dual-use markets beyond those already present in international trade generally.

Next, we examine how the intensity and composition of export activity subject to the EAR varies across firm-size categories to understand whether compliance costs include substantial fixed components that may create competitive disadvantages for smaller exporters. We restrict our sample to firms with at least one export shipment subject to the EAR and divide firms into four size bins, the smallest being firms with fewer than 50 employees and the largest having 500 or more employees.

In the first column of Table 3 we show the average share of firm-level export value that is accounted for by shipments subject to the EAR. Conditional on the firm having at least one shipment subject to the EAR, 63% of the average small firm's exports, by value, are accounted for by controlled products, compared to only 53% of the average large firm's export value. This suggests that small firms with exports subject to the EAR are more likely to specialize in controlled product exports, whereas large firms export controlled products as part of a more diversified portfolio.

The second column shows, for each size bin, the fraction of firms that had at least one shipment with an ECCN. (As is the case in the first column, the sample includes only firms with at least one shipment subject to the EAR.) Among small firms, only 11% exported any ECCN, implying that 89% exported only EAR99-designated shipments. This share rises monotonically with firm size: 34% of large firms exported at least one shipment with an ECCN. The relative specialization of small firms in controlled products and their lower likelihood of exporting shipments requiring ECCNs could be due to products with ECCNs being more complex and costly, requiring a larger scale to produce; or it may reflect an additional regulatory burden associated with these exports.

The next two columns show the average number of unique ECCNs per firm, and, for context, the average number of six-digit Schedule B codes per firm. There are approximately 400 unique ECCNs in total (see Appendix A). In contrast, there are more than 5,600 unique six-digit Schedule B codes, so we expect the average number of Schedule B codes per firm to be larger. Small firms average just 1.1 ECCNs (counting EAR99 as an ECCN), compared to 2.5 for large firms. Similarly, small firms export an average of 5.7 distinct six-digit Schedule B codes, compared to 54.3 for large firms—a tenfold difference. These patterns reinforce the specialization story: small firms concentrate exports subject to the EAR in a narrow range of products, whereas large firms maintain diversified portfolios spanning many product categories.

The Role of Multinational Firms

Beyond firm size, multinational status may affect a firm's ability to navigate export controls. Multinational entities (MNEs) can structure cross-border transactions as intra-firm trade between affiliated entities rather than through arm's-length market transactions. Standard trade theory suggests several reasons for internalization, including mitigating risks of technology leakage and incomplete contracting when transferring proprietary knowledge (Antràs, 2003). In the context of export controls, intra-firm transactions may also offer organizational advantages: MNEs possess established centralized compliance structures, superior capabilities for monitoring end-use, and internal oversight to meet complex regulatory requirements. If these organizational capabilities provide distinct advantages in managing dual-use export compliance, we should expect controlled products to exhibit higher shares of intra-firm trade. Table 4 reports the prevalence of related-party transactions across product and destination categories.

Two key patterns emerge. First, across all destinations, trade in EAR-controlled products displays a substantially higher related-party share (42%) than overall U.S. exports (roughly one-third), indicating that trade in controlled goods is disproportionately conducted within multinational boundaries. Second, across all product categories, related-party trade shares are systematically lower for exports to high-risk destinations (Country Group D, where related-party trade falls to 20% overall and 30% for EAR goods) compared to lower-risk destinations (Country Group B).

The lower related-party share of controlled exports to Group D destinations primarily reflects the lower baseline density of U.S. multinational affiliate networks in these more restricted destinations. Conditional on operating in these markets, however, MNEs exporting EAR-controlled goods continue to rely more heavily on intra-firm channels (30%) than non-controlled exporters (20%). For products with specific ECCNs, related-party shares mirror overall export averages, suggesting that while MNE organizational advantages are pronounced across EAR-controlled goods generally, ECCN items follow broader baseline patterns of multinational trade structure.

The Role of R&D Performers

A final dimension of heterogeneity we explore is firms' engagement in R&D activities. We examine whether R&D-performing firms, which tend to be larger, account for a disproportionate share of exports subject to the EAR. This analysis speaks to concerns that export controls may particularly constrain innovative firms that depend on global markets to recoup substantial R&D investments. In the BERD, we distinguish between known R&D perform-

ers, on the one hand—firms that responded to the survey and provided a nonzero dollar investment amount—and all other firms. The latter category may include R&D performers, if they were not included in the BERD sample or did not respond to the survey. Known R&D performers account for 3.3% of all firms with export shipments in the 2023 LFTTD (4,600 of 191,400), and, as Table 5 shows, for 55% of all export value. They also account for 65% of the value of exports subject to the EAR. The higher prevalence of controlled compared to overall exports at R&D performing firms likely reflect both the technological content of their products as well as their ability to manage compliance costs given they are also highly engaged in exporting overall.

The portrait of dual-use exporters reveals that export-control exposure is highly concentrated among large firms. Because these large firms hold extensive product portfolios and global supply networks, policy shocks targeting specific dual-use goods may induce broader spillovers across their non-targeted business lines. To rigorously estimate the direct and indirect causal impacts, we use a natural experiment from the 2014 U.S. sanctions on Russia in the next section.

5 U.S. Exporter Responses to 2014 Russia Sanctions

Background

In February 2014, Russia invaded and began occupying Crimea, prompting the U.S. and its Western allies to impose sanctions on Russia, including trade restrictions (Reuters, March 8, 2014). The 2014 U.S. sanctions administered under the BIS included the addition of Russian individuals and companies to the Entity List and the designation of specific ECCN and Schedule B codes subject to export controls. We study the impacts of the regulations in FR 45675 effective August 6, 2014, which added eight ECCNs and 52 Schedule B codes to the list of exports to Russia subject to licensing requirements with a presumption of license denial.[15] We refer to these collectively as the targeted products. More information on the Russia-targeted sanctions is in Appendix E.

[15]We focus on the 2014 sanctions on Russia rather than the much larger set of sanctions imposed in 2022 following Russia's invasion of Ukraine because, unlike the 2014 sanctions, which were narrowly targeted, the trade sanctions imposed following Russia's second invasion of Ukraine in February 2022 have been much more expansive and very dynamic: there have been multiple rounds of sanctions, including new licensing requirements, new prohibitions targeting oil exploration, and sanctions on specific Russian firms and individuals. The ratcheting-up nature of the 2022 sanctions make it difficult to study the impact of any individual restriction on trade.

These export controls had an immediate effect on export-license issuance. Figure 2(a) shows the number of new licenses granted by the BIS for export of tangible items, software, and technology to Russia. The number of approved license applications plummeted from over 1,800 in 2013 to a little over half of that number in 2014 (with sanctions imposed midyear), to around 250 per year from 2015 to 2017, and just over 100 per year in 2018 and 2019.[16]

Figure 3(a) shows the real export value of targeted products (solid line) and exports of all other products (dashed line) to Russia starting in the first quarter of 2012, just before Russia's World Trade Organization accession in August 2012, to the second quarter of 2018. There was a marked decline in exports of both targeted and non-targeted goods.[17] However, the export value of targeted products does not fall to zero. We believe this likely reflects our inability to observe end users, as a result of which some transactions that do not require licenses may be misclassified as targeted transactions. These patterns suggest the sanctions created a broad chilling effect that dissipated for non-targeted products but persisted for targeted goods.

Samples and Empirical Approach

Treatment Sample

We identify treated firms that were subject to the Russia-targeted export controls if they exported a product on the targeted list (based on ECCN, Schedule B, or both) to Russia in 2012 and/or 2013.[18] This is the pre-treatment period. There are approximately 550 firms affected by the Russia-targeted export controls.[19]

An observation in our analysis is a firm-year. We aggregate to the year level because transactions are likely to be "lumpy," with a large shipment in one month and no transactions in other months (Crozet and Hinz, 2020, p. 106; see also Figure 3 in this paper). Because the Russia-targeted export controls were imposed in the middle of 2014, and because

[16]The approved license applications are drawn from BIS's annual report to the U.S. Congress that are filed on a fiscal-year basis. However, the annual reports in 2018 and 2019 do not provide statistics on approved applications. For these years, we rely on the annual country-licensing and trade analysis that reports approved license applications on a calendar-year basis (U.S. Department of Commerce, Bureau of Industry and Security, 2022). Therefore, in Figure 2, years 2012–2017 refer to fiscal years, and years 2018–2019 refer to calendar years.

[17]The spike in the value of targeted exports apparent in the second quarter of 2016 is due to four shipments, totaling over $140 million in 2019 dollars, of Schedule B code 8430498010: "Offshore oil and natural gas drilling and production platforms for oil well and gas field drilling, rotary." These shipments do not have an ECCN but are tagged as EAR99 in U.S. Census Bureau (2024).

[18]Targeted ECCNs include software that may not be captured in merchandise trade transactions. Appendix E provides an aggregate analysis of exports of targeted services product categories to Russia.

[19]All counts are rounded to comply with Census Bureau disclosure-avoidance rules.

of the possibility of anticipatory effects during Russia's amassing of troops on the border with Crimea and during the invasion and annexation, but before the Russia-targeted export controls were imposed, we drop the period from January through July of 2014 from our analysis. Consequently, post-treatment years for the treated firms run from August through July of the following calendar year.

Table 6 shows summary statistics for the treated firms in the pre-treatment period.

Control Sample

We identify control firms as those that exported the same set of targeted products to China—another Group D country—in the same time period, excluding any firms that are in the treatment sample. Because China is a large market, we find a sufficiently large control sample: about 1,100 firms.

The bottom panels of Figures 2 and 3 show how licenses and exports of targeted and non-targeted products to China evolved over the period 2012–2019. More than 2,000 export-license applications of controlled products to China were approved in 2012, somewhat more than were approved to Russia that year. In stark contrast to the pattern of approved applications to Russia, the number of approved licenses for export to China rose steadily in subsequent years, increasing to nearly 3,000 by 2017. The real export value of targeted products to China tripled from 2012 to 2015, then fell back to 2012 levels by 2019.

Empirical Approach

We estimate difference-in-differences (DD) regressions that compare the outcomes of treated firms—those that exported targeted products to Russia before the Russia-targeted export controls took effect—to outcomes of control firms that exported the same set of products to China in the same time period. The DD specification is:

$$\text{Outcome}_{it} = \alpha_i + \delta_t + \beta \cdot \text{Treated}_t \times \text{Post}_i + \varepsilon_{it}. \tag{1}$$

Treated indicates that firm i is in the treated sample. The coefficient β measures the difference in the outcome variable for an average treated firm between the period after the Russia-targeted export controls were implemented in 2014 and the period before 2014, compared to the difference in the outcome variable for the average control firm. The main identifying assumption is that, but for the imposition of the Russia-targeted export controls, the exports of firms selling to Russia would have followed the same trend as the exports of firms selling to China.

We examine both extensive and intensive margins of adjustment. Extensive-margin outcomes are of the form $\mathbb{I}(\text{Exports}_{it})$, where $\mathbb{I}(\cdot)$ is an indicator that equals 1 if firm i has any exports in year t; intensive-margin outcomes are of the form $\ln(\text{Exports}_{it})$, and omit all zero values. We examine the value of exports of targeted products to the focal country, the value of exports of non-targeted products, the value of exports to other destinations, and the value of all exports by firm i. The focal country is Russia for the treated firms and China for the control firms.

Results

We begin by examining whether a treated firm's probability of exporting targeted goods to Russia decreased following the Russia-targeted export controls in the first column of the top panel of Table 7. The estimated coefficient indicates a decrease of about 8.5 percentage points (pp). This estimate captures the direct effect of the Russia-targeted export controls and establishes a necessary first-stage impact to guide subsequent analysis. To evaluate the economic significance of this decline, note that the probability of exports of targeted products to the focal country falls, mechanically, for both the treated and the control firms starting in 2014. This is because both treatment and control firms are selected into our sample with the condition of having some exports of targeted products to the focal country in the pre-treatment period. Over time, firms may exit export markets due to changes in their export-product mix and destination countries, or they may go out of business altogether. The probability that the firms in the control group export targeted products to China declined from 64.5% in 2012 and 2013 to 24.7%, on average, in the post-treatment years, a drop of nearly 40pp. We interpret the additional decline of 8.5pp among the treated firms in the probability of exporting targeted products to Russia as a consequence of the export controls.[20]

Exports of targeted products to Russia could have fallen due to compliance with the Russia-targeted export controls, but they could also have fallen because firms either stopped reporting ECCNs on their export forms or diverted shipments to other destinations, from which they may be smuggled into Russia. The next two columns of Table 7 help rule out these scenarios. They show the impact of the Russia-targeted export controls on the probability that a firm exported non-targeted products to any destination (second column) and the probability that it exported any products to the non-focal destination (third column). The

[20]This result implies that some targeted shipments to Russia continued after the imposition of the Russia-targeted export controls, consistent with aggregate flows of targeted products to Russia, which show sustained declines after 2014 but not a complete halt (Figure 3(a)).

coefficients are all small, negative, and statistically insignificant. Together, we take this as evidence that it is unlikely that treated firms *increased* their probability of exporting non-targeted products, nor their probability of exporting to other destinations, following the imposition of the Russia-targeted export controls. The fourth column shows our estimates of the impact on a firm having any exports (of any product, to any destination). The DD coefficients on both these outcomes are statistically insignificant, indicating that the sanctions did not have a differential impact on exit from export markets or exit from business altogether.

In the bottom panel of Table 7, we replace the binary outcome variables with their corresponding log value, restricting the sample to observations with non-zero exports in each case. We do not estimate the intensive margin of exports of targeted products to the focal country because there are so few firms, in both the treated and control sample, that have positive exports of these products in the later years. For the other outcomes, the intensive-margin responses are economically large and statistically significant. The DD estimates imply that, conditional on continued exports, treated firms experienced, on average, 20% decrease in total export value. At the pre-treatment mean of total export value for treated firms, $376 million, this implies a decline of approximately $75 million for the average treated firm relative to the pre-sanctions period. We estimate similar percentage-rate impacts on the export value of non-targeted products and export value of shipments to destinations other than the focal country. Together, the results imply that the sanctions' impact extended far beyond the directly restricted transactions. These large decreases across multiple margins suggest that firms are, by and large, not simply diverting shipments to other destinations or misreporting ECCNs on their customs forms. Instead, the policy resulted in real disruptions to firm operations, with compliance costs, reputational concerns, or production complementarities driving reductions in economic activity beyond what the export controls explicitly targeted.

To examine the impact of the export controls on domestic operations, the last column uses LBD payroll. In the top panel, we have an indicator for the firm having any activity in the LBD; in the bottom panel, we use the log of payroll as a measure of activity. The probability that a treated firm has positive payroll does not change differentially for the treated and control firms in the aftermath of the Russia-targeted export controls—the coefficient of 0.0043 is small in magnitude and statistically insignificant, indicating that firms did not exit the market entirely in response to the sanctions. Conditional on the firm having positive payroll, however, the payroll declines by approximately 11% on average among the treated firms relative to the control firms. At the mean payroll of $254 million for treated firms in the pre-period, this represents a reduction of roughly $27 million per firm.

This substantial payroll decline, combined with the absence of excess exit, suggests that

affected firms responded to the export controls through workforce reductions and operational downsizing rather than complete shutdown. The magnitude of the payroll response indicates that the costs of export controls are not borne solely through lost international sales but also translate into economically significant domestic employment effects. This finding underscores that restrictions on exports can negatively impact domestic workers and production capacity, even when firms remain operational.

6 Concluding Remarks

Export controls have become a central tool in U.S. trade policy, but an assessment of the costs they impose on domestic firms has been challenging due to limitations in data availability of the distinct product classification system used to administer controlled products. Export controls can impose compliance burdens, create market uncertainty, and disadvantage domestic firms relative to foreign competitors that may face fewer restrictions. Without empirical evidence on how export controls affect firm behavior it is difficult to evaluate whether alternative policy approaches—such as more targeted restrictions, enhanced screening mechanisms, or international coordination—could achieve similar security objectives at lower economic cost.

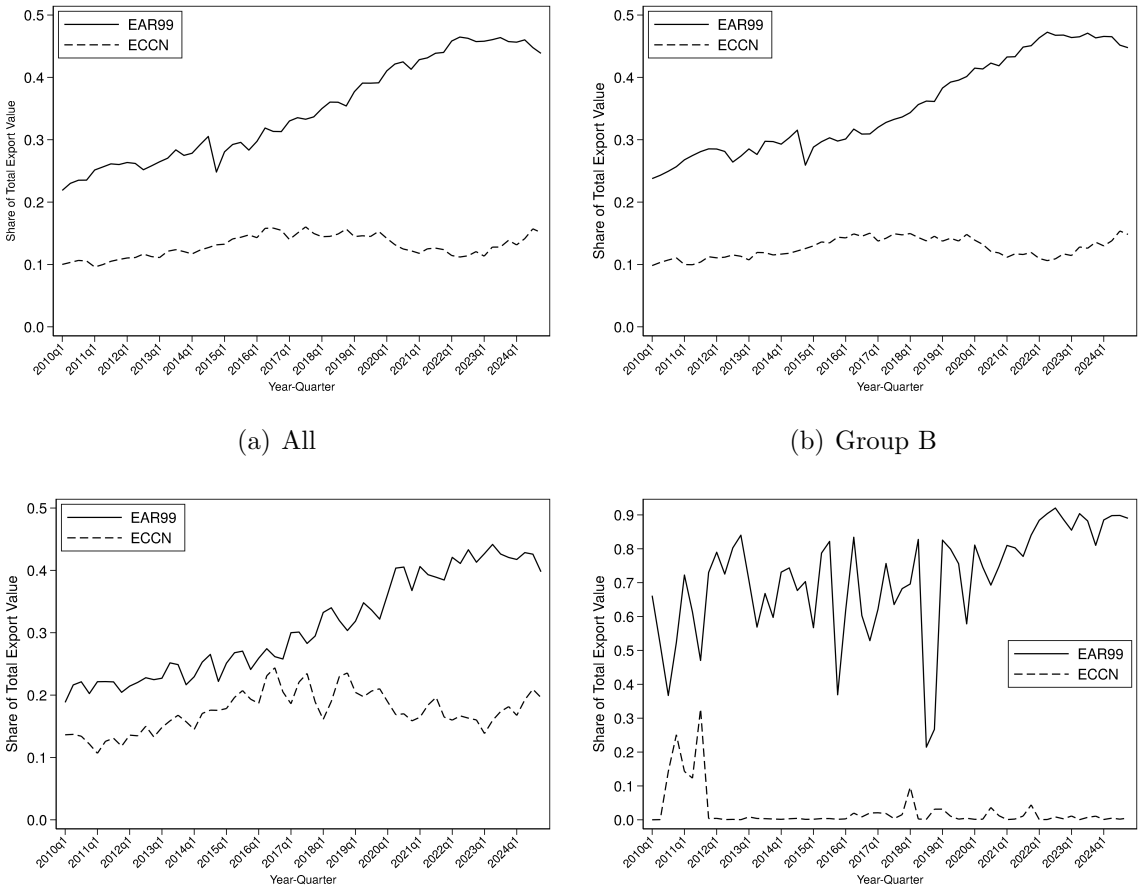
Using newly available public export statistics containing product codes used in the U.S. export control system, ECCNs and EAR99, combined with confidential firm-level customs transactions, we present three main results. First, the scope of export controls has expanded dramatically since 2010: the share of U.S. exports subject to controls doubled between 2010 and 2024, driven primarily by products designated EAR99. Second, although exports subject to the EAR are primarily supplied by large firms, small exporters that export controlled products are significantly more specialized in controlled products, particularly those designated EAR99. This pattern is consistent with dual-use market participation being shaped by both compliance costs and more complex production processes associated with ECCNs. Third, examining the 2014 episode of U.S. sanctions on Russia, we find that in addition to reducing U.S. firms' exports of targeted products to Russia, the export controls also reduced their exports of products and to destinations not directly targeted by controls and lowered domestic payroll.

The substantial spillover effects we observe—declines in exports of non-targeted products and to non-Russia destinations—can arise through several channels. Production complementarities may link targeted and non-targeted products through shared inputs, manufacturing facilities, or technical expertise. When firms lose access to the Russia market for certain products, scale effects may reduce the profitability of their entire product portfolio, leading

them to curtail production more broadly. In addition, export controls may impose firm-wide compliance costs that extend beyond the specific restricted transactions. Firms must invest in legal review, maintain detailed documentation systems, implement monitoring protocols, and potentially restructure their operations to ensure compliance. These fixed costs may deter firms from exporting to other sensitive destinations where similar scrutiny might apply, effectively creating a chilling effect on trade beyond what the regulations explicitly target. Reputational concerns and policy uncertainty can amplify the direct impact of controls. Downstream buyers may avoid suppliers with export-control exposure due to concerns about supply-chain reliability or their own compliance risks. Additionally, uncertainty about future restrictions—particularly given the rapid evolution of export-control policy in recent years—may lead firms to reduce investment in export capacity more generally, viewing international markets as increasingly risky. Together, these mechanisms imply that the economic costs of export controls can extend substantially beyond the value of directly restricted transactions.

Although a complete welfare assessment, which would necessarily include a quantification of national-security benefits, is beyond the scope of this paper, our evidence indicates the importance of accounting for indirect costs when evaluating the economic consequences of export controls. As export controls continue to expand in scope and application, further research quantifying these tradeoffs will be valuable for informing policy design that balances national-security objectives with economic considerations.

Figure 1. Controlled Exports as Share of All U.S. Exports
All and by Country Group, 2010–2024



(a) All

(b) Group B

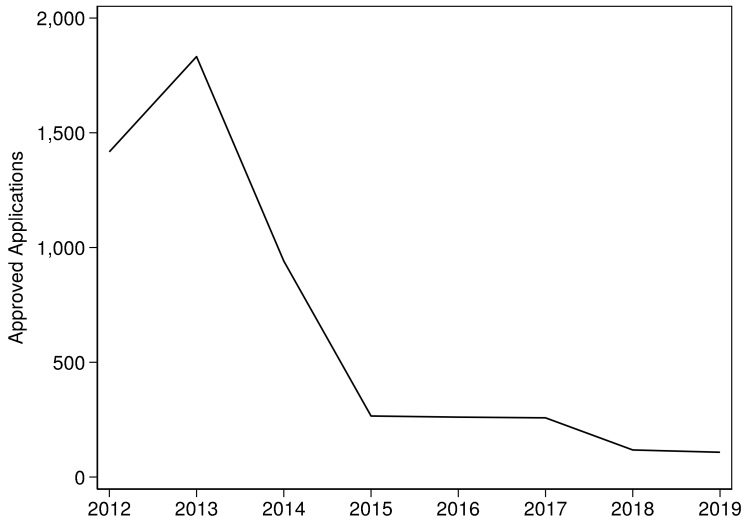
(c) Group D

(d) Group E

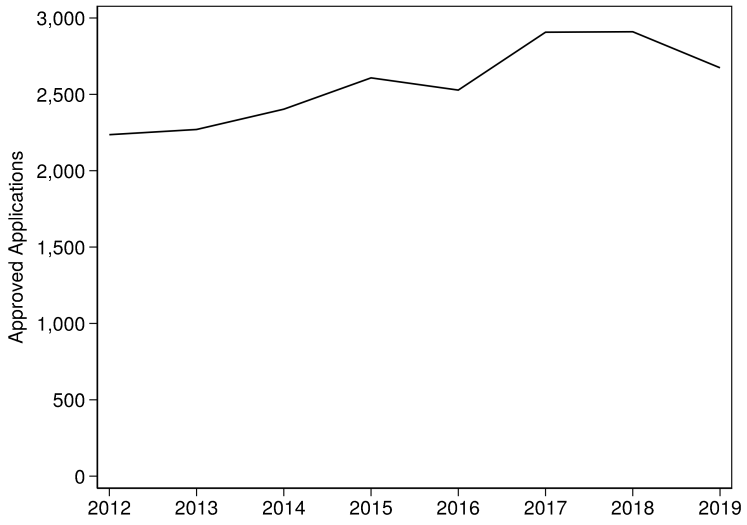
Notes: Value accounted for by shipments designated as EAR99, and shipments designated ECCNs, as a share of all U.S. exports, by country group (defined in Section 2). ECCN designates shipments that are subject to the EAR and are listed in the CCL. EAR99 designates shipments that are subject to the EAR but are not listed in the CCL. Calculations exclude exports to Canada, Puerto Rico, and the U.S. Virgin Islands.

Source: Authors' calculations using U.S. Census Bureau (2024) and country group designations from U.S. Department of Commerce, Bureau of Industry and Security (2025j).

Figure 2. Number of License Applications Approved for Exports, 2012–2019 by Destination Country



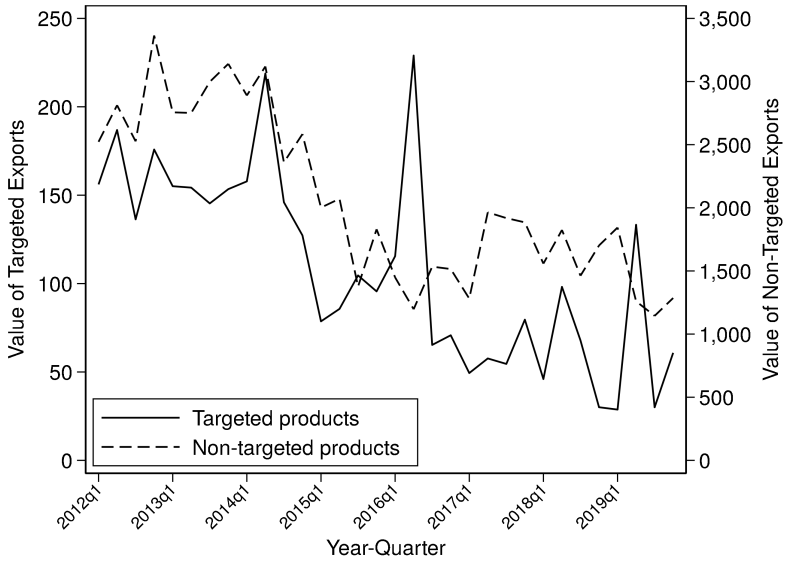
(a) Russia, 2012–2019



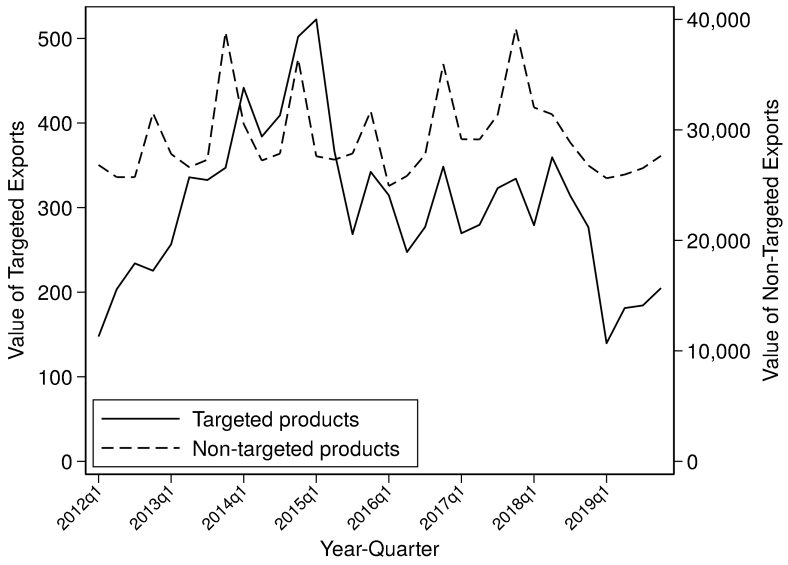
(b) China, 2012–2019

Notes: Number of approved license applications for dual-use exports to (a) Russia and (b) China. Excludes "deemed" licenses (a small category including university research).
Source: Authors' calculations using Bureau of Industry and Security Annual Country Licensing and Trade Analysis for a calendar year (U.S. Department of Commerce, Bureau of Industry and Security, 2025a).

Figure 3. Targeted vs. Non-Targeted Product Exports to Russia and China



(a) Russia



(b) China

Notes: Value of targeted and other products, in millions of 2019 U.S. dollars, exported to (a) Russia and (b) China. Targeted products are defined as one of 52 Schedule B codes or eight ECCN codes listed in FR 45675; non-targeted products refer to all other products not listed in FR 45675.
Source: Authors' calculations using U.S. Census Bureau (2024) and FR 45675.

Table 1. Acronyms Used in the Paper

Acronym	Full Name or Definition	Page[a]
AES	Automated Export System	8
BERD	Business Enterprise Research and Development Survey	8
BIS	Bureau of Industry and Security	4
CCL	Commerce Control List	4
CN	Common Nomenclature	26
DD	Difference in difference	14
EAR	Export Administration Regulations	4
EAR99	Product subject to EAR but not on CCL	4
ECCN	Export Control Classification Number	4
EEI	Electronic Export Information	33
FR 45675	Federal Register notice 45675: U.S. Department of Commerce, Bureau of Industry and Security (August 6, 2014)	2
LBD	Longitudinal Business Database	8
LFTTD	Longitudinal Firm Trade Transactions Database	7
MNE	Multinational Enterprises	11
R&D	Research and development	8
pp	Percentage points	15
Schedule B	10-digit export product classification	8

[a] Page number in text for more information.

Table 2. Firm Concentration in Controlled Exports

	Share of Export Value	
Export Type	Top 1%	Top 10%
All U.S. Exports	0.8396	0.9723
Exports subject to the EAR	0.8763	0.9791
ECCN Exports	0.8247	0.9781

Notes: Sample of 191,400 firms with at least one export shipment in the 2023 LFTTD. Top 1% and top 10% firms are defined separately for each export type. All calculations exclude exports to Canada.
Source: Authors' calculations from 2023 LFTTD.

Table 3. Controlled Exports by Firm Size

Firm Employment	% Export Value Subject to EAR	% Firms Exporting ECCN (non-EAR99)	ECCNs per Firm (including EAR99)	Six-digit Schedule Bs per Firm
<50	63.3	11.0	1.114	5.7
50–249	54.8	13.3	1.197	9.6
250–499	52.0	18.6	1.392	17.8
≥500	53.1	34.3	2.514	54.3

Notes: Sample of 79,400 firms that have at least one 2023 shipment subject to the EAR and have nonmissing 2023 LBD employment. Employment bins are based on March 12, 2023 employment in the LBD. Statistics reflect firm averages. The calculation of ECCNs per firm treats EAR99 as an ECCN. All calculations exclude exports to Canada.
Source: Authors' calculations from 2023 LFTTD and 2023 LBD.

Table 4. Related-Party Exports in Controlled Products

	% Related-Party Export Value to...		
Export Type	All Destinations[a]	Group B	Group D
All U.S. Exports	32.9	35.8	20.6
Exports subject to the EAR	41.8	44.8	30.5
ECCN Exports	31.4	35.3	20.2

Notes: Sample includes all transactions in 2023 LFTTD with non-missing value for related-trade indicator, excluding exports to Canada. Related-party trade is defined as transactions between entities with common ownership. Country groups are assigned based on a country's "worst" grade in U.S. Department of Commerce, Bureau of Industry and Security (2025c). Each cell represents related-party exports of a given type (row) to a set of destinations (column) as a share of all exports of that type to that set of destinations.
[a] "All destinations" includes countries in Group E as well as countries with no country group, but excludes Canada.
Source: Authors' calculations from U.S. Census Bureau (2025b) for all U.S. exports; authors' calculations from 2023 LFTTD for EAR99 and ECCN exports.

Table 5. Controlled Exports by R&D Activity

R&D Status	% of All Export Value	% Export Value Subject to EAR
Known R&D performers	55	65

Notes: Statistics are aggregates, not firm averages. Sample of 191,400 firms with at least one export shipment in the 2023 LFTTD. Known R&D performers responded to the 2023 BERD survey with a positive value of R&D expenditures. All other firms, including those with unknown status and firms known to be non-R&D peformers, are in the omitted category. All calculations exclude exports to Canada.
Source: Authors' calculations from 2023 LFTTD and 2023 BERD.

Table 6. Summary Statistics, Treated Firms,
Pre-Treatment Period

	Mean	Std Dev
Prob(Exported targeted products to Russia)	0.684	
Export value, targeted products to Russia[a]	1.568	7.164
Exports of targeted products to Russia as a share of firm's export portfolio	0.123	
Prob(Any Exports)	0.962	
Export value[a]	376	1,970
Prob(Exported non-targeted products to any destination)	0.922	
Export value, non-targeted products to all destinations[a]	371	1,970
Prob(Exported to destinations other than Russia)	0.891	0.864
Export value, all products to destinations other than Russia[a]	400	2,020
Payroll[a]	254	994

Notes: Annual mean and standard deviation (for continuous variables) for 550 treated firms the pre-treatment period. Treated firms exported at least one of the targeted products to Russia in 2012 or 2013. All statistics are rounded to comply with Census Bureau disclosure-avoidance rules.

[a] Conditional on nonzero value; expressed in millions of 2019 U.S. dollars.

Source: Authors' calculations using LBD, LFTTD, and FR 45675.

Table 7. Impact of Russia-Targeted Export Controls on the Extensive and Intensive Margins of Firm Exports

	Probability of...				
	Exports of Targeted Products to Focal Country[a]	Exports of Non-Targeted Products	Exports to Destinations Other than Focal Country[a]	Any Exports	Any Activity[b]
Treated × Post	−0.0855***	−0.0212	−0.0235	−0.0094	0.0043
	(0.0153)	(0.0157)	(0.0152)	(0.0156)	(0.0173)

	Log of...[c]			
	Exports of Non-Targeted Products	Exports to Destinations Other than Focal Country[a]	Exports	Payroll
Treated × Post	−0.2266***	−0.2132***	−0.2816***	−0.1117***
	(0.0701)	(0.0688)	(0.0666)	(0.0371)

Notes: Coefficients from difference-in-difference regressions, defined in Equation (1), on binary variables (top panel) and continuous variables (bottom panel). Robust standard errors in parentheses, clustered by firm. All regressions include year and firm fixed effects.

[a] Focal country is Russia for the treated sample and China for the control sample.

[b] Indicator for any domestic payroll in LBD.

[c] Includes only observations with non-zero values.

* significant at 10%; ** significant at 5%; *** significant at 1%

Source: Authors' calculations using LBD, LFTTD, and FR 45675.

Supplementary Appendix

A Relationship between ECCN and Schedule B Codes

f An export shipment subject to export controls under EAR contains two distinct product classifications: a Schedule B code and an ECCN. U.S. exporters must include a Schedule B code—a ten-digit number used to classify goods for exports from the U.S.—when reporting their export shipments to U.S. Customs and Border Protection. These codes are maintained by the U.S. Census Bureau (U.S. Census Bureau, 2025c). When appropriate, exporters must also list an ECCN on their export filings. ECCNs are determined based on the use of the product, whereas Schedule B codes are based on the product's physical characteristics. Thus, there does not exist a one-to-one mapping between the Schedule B and ECCN classifications. A single ECCN may map into multiple Schedule B codes, and one Schedule B code may map into multiple ECCNs. The same Schedule B code may also be used in shipments that have no listed ECCN.

The first character of the ECCN, a number, identifies the broad category to which it belongs, such as nuclear materials, electronics, or sensors and lasers. The second character identifies the product group, such as test equipment or software. The last three digits indicate the reason for control, such as national security or nuclear nonproliferation. Table A-1 shows that Aerospace and Propulsion ECCNs account for approximately half of controlled export value excluding EAR99, followed by Telecommunications and Information Security, and then Electronics. Category shares remain stable between 2010 and 2024, except for a slight shift from Electronics toward Telecommunications.

The many-to-many relationship between ECCN and Schedule B code underscores the challenge in precisely identifying controlled shipments without information on export flows by ECCN. Researchers often use Schedule B codes to proxy for ECCNs. For example, Sundaram and Richardson (2013, Table 2) identify exports subject to controls as those in seven broad high-technology sectors. In recent work, Liu, Liu, and Wen (2025) use a correlation table published by the European Commission to convert ECCNs to six-digit Schedule B codes to study the impact of U.S. export controls on Chinese firms.[21] Kohl, van den Berg, and Franssen (2023) use CN8 codes to identify dual-use products. Although the U.S. Census Bureau has only recently started publishing data on ECCN exports (U.S. Census Bureau, 2024), the BIS has been explicit about the lack of correspondence: "Schedule B numbers are not related to Export Control Classification Numbers" (U.S. Department of Commerce, Bureau of Industry and Security, 2025g, p. 1). Thus, any mapping between standard traded-product and dual-use product classifications is necessarily incomplete, and may result in both under- and over-classification of exports into the controlled set.

[21] The European Commission provides a correlation between eight-digit Common Nomenclature (CN8) and dual-use codes, which demonstrates the many-to-many mapping between the two classification systems (European Commission, 2025). Chatelus and Heine (2016) find that about a third of correlations are likely incorrect in the EU correlation system. ECCN is similar when a product appears in both the U.S. CCL and the EU dual-use list. See Bromley and Brockmann (2024) for a discussion of the differences between the U.S. and EU export-control systems.

To illustrate the many-to-many relationship between Schedule B codes and ECCNs, Table A-2 shows a mapping of Schedule B codes to ECCNs in 2024 based on U.S. Census Bureau (2024). In 2024, across all shipments, there were 9,407 unique ten-digit Schedule B codes used. Over 90% of Schedule B codes were associated with at least one shipment that was designated as EAR99 or included an ECCN; most of these were designated EAR99, but about 40% of all ten-digit Schedule B codes were associated with at least one ECCN, and 5% were associated with at least 11 ECCNs. At the extreme, Schedule B code 9031808080, which includes "measuring or checking instruments, appliances and machines, not specified or included elsewhere in this chapter; profile projectors; parts and accessories thereof" was associated with 85 unique ECCNs. Both Schedule B codes and ECCNs change over time, and the relationship between them is also dynamic: for example, Schedule B codes associated with an ECCN in 2024 are not necessarily associated with the same ECCN (or any ECCN) in other years.

Table A-3 reverses the lens and analyzes the ECCNs in the 2024 data. Of the nearly 400 unique ECCNs in the data, 52 were used to tag a single shipment, and another 20 were associated with a single Schedule B code across multiple shipments. The vast majority of ECCNs were used with multiple ten-digit Schedule B codes. In fact, 300 ECCNs—three quarters of all ECCNs—were associated with more than one two-digit Schedule B code (more than one chapter), and 18.5% were associated with more than ten chapters. At the extreme is ECCN 9A991 (aircraft parts and components). The vast majority of exports with ECCN 9A991 have a Schedule B code in chapter 88 (aircraft, spacecraft, and parts thereof); however, it is associated with Schedule B codes in 77 different chapters, including codes in chapter 28 (chemical products), chapter 39 (articles from rubber and plastic), and chapter 51 (wool, fine or course animal hair; horsehair yarn and woven fabric); and with over 1,000 unique ten-digit Schedule B codes.

When designating products subject to national-security-based export controls, BIS may use ECCNs alone or in combination with Schedule B codes.[22] For example, U.S. sanctions on Libya in 2004 were entirely ECCN based: the BIS announced a policy of denial of export licenses of certain ECCNs to Libya, and a review of applications for export licenses of other specified ECCNs on a case-by-case basis (U.S. Department of Commerce, Bureau of Industry and Security, April 29, 2004). In contrast, when U.S. sanctions on Russia were expanded to include Belarus in 2024, the BIS announced restrictions not only on specific ECCNs, but also on a long list of Schedule B codes (U.S. Department of Commerce, Bureau of Industry and Security, June 18, 2024). These examples underscore the importance of having ECCN information alongside Schedule B codes to study the impact of export controls.

[22]For a list of all regulations related to export controls that have been issued by BIS, see U.S. Department of Commerce, Bureau of Industry and Security (2025e).

Table A-1. U.S. Controlled Export Value and Share by CCL Category

Controlled Category	Description	2010 Value[a]	2010 Percent[b]	2024 Value[a]	2024 Percent[b]
0	Nuclear Materials	490	0.1	1,434	0.2
1	Chemicals and Toxins	2,476	0.6	3,444	0.4
2	Tools for Materials Processing	2,208	0.5	4,188	0.5
3	Electronics	23,888	5.5	22,014	2.5
4	Computers	10,674	2.5	17,547	2.0
5	Telecomm and Information	30,129	7.0	49,670	5.7
6	Sensors and Lasers	2,263	0.5	1,590	0.2
7	Navigation and Avionics	2,752	0.6	4,879	0.6
8	Marine	1,083	0.3	1,073	0.1
9	Aerospace and Propulsion	59,062	13.7	106,845	12.3
	EAR99	296,566	68.7	659,260	75.6
	TOTAL	431,591		871,946	

Notes: Export values for 2010 and 2024 of all shipments associated with an ECCN or EAR99. "Controlled Category" refers to the first digit of an ECCN.
Source: Authors' calculations using U.S. Census Bureau (2024). Category descriptions are based on U.S. Department of Commerce, Bureau of Industry and Security (2025b).
[a] Millions of 2019 dollars.
[b] May not sum to 100% due to rounding.

Table A-2. Mapping of Schedule B Codes to ECCNs, 2024

	Count	Percent[a]
Unique ten-digit Schedule B codes	9,393	100.0
Schedule B codes never associated with EAR99 or an ECCN	608	6.5
Schedule B codes associated with EAR99 but no ECCN	5,038	53.6
Schedule B codes associated with exactly one ECCN	1,341	14.4
Schedule B codes associated with 2-10 ECCNs	1,900	20.2
Schedule B codes associated with 11 or more ECCNs	506	5.4

Notes: Mapping from Schedule B Codes to ECCNs using export shipments in 2024. Each shipment is associated with a ten-digit Schedule B code and, if applicable, EAR99 or an ECCN. We omit three Schedule B codes from this analysis: 0000000001 (invalid code), 8800000000 (suppression code used by the Census Bureau; see U.S. Census Bureau, 2009), and 9801100000 (value of repairs or alterations of previously imported articles, repaired or altered prior to exportation from the U.S.).
Source: Authors' calculations using U.S. Census Bureau (2024).
[a] May not sum to 100% due to rounding.

Table A-3. Mapping of ECCNs to Schedule B Codes, 2024

	Count	Percent[a]
Unique ECCNs (excluding EAR99)	399	100.0
ECCNs associated with a single shipment	52	13.0
ECCNs associated with more than one shipment but with only one ten-digit Schedule B code	20	5.0
ECCNs associated with more than one ten-digit Schedule B code, but within a single four-digit Schedule B code	10	2.5
ECCNs associated with more than one four-digit Schedule B code, but within a single two-digit Schedule B code	17	4.3
ECCNs associated with 2-10 two-digit Schedule B codes	226	56.6
ECCNs associated with 11 or more two-digit Schedule B codes	74	18.5

Notes: Mapping from ECCNs to Schedule B Codes using export shipments in 2024. Each shipment is associated with a ten-digit Schedule B code and, if applicable, EAR99 or an ECCN. Three Schedule B codes are omitted from this analysis: 0000000001 (invalid code), 8800000000 (suppression code used by the Census Bureau; see U.S. Census Bureau, 2009), and 9801100000 (value of repairs or alterations of previously imported articles, repaired or altered prior to exportation from the U.S.).
Source: Authors' calculations using U.S. Census Bureau (2024).
[a] May not sum to 100% due to rounding.

B Control Country Groups

As noted in Section 2, licensing regulations vary by country group (U.S. Department of Commerce, Bureau of Industry and Security, 2025c). Some countries may be in multiple groups because license exceptions are triggered by the combination of country, product (ECCN), and end user/use.

Group A countries form a strict subset of Group B countries. These countries are singled out for the most favorable treatment because they participate in a multilateral export-control regime with the United States.[23] The multilateral regimes are indicated by the sub-groups under Group A as follows: A:1 Wassenar participating states; A:2 missile technology control regime; A:3 Australia group; and A:4 nuclear suppliers group. Subgroups A:5 and A:6 group countries that are eligible for license exceptions under the Strategic Trade Authorization (U.S. Department of Commerce, Bureau of Industry and Security, 2025i). Groups A:5 and A:6 differ by the type of products eligible for exceptions. Countries in Group A may be listed in multiple subgroups; unlike the group classifications, the subgroups do not convey a hierarchy of control stringency.

Most countries are in Group B. This is the default country group, and it includes all Group A countries.

There are five sub-groups under Group D, four of which denote specific proliferation concerns; the last sub-group denotes an arms-embargo status. The sub-groups are as follows: D:1 National Security; D:2 Nuclear; D:3 Chemical and Biological; D:4 Missile Technology; and D:5 U.S. Arms Embargoed Countries. As with Group A, countries may be in multiple subgroups, which are not hierarchical.

There are only four countries in Group E: Cuba, Iran, North Korea, and Syria. Group E countries are never in any other groups.

[23] Fan, Handley, and Limão (2026) find that accession to specific multilateral export-control regimes is associated with increases in bilateral trade between member countries relative to nonmembers.

C Patterns in Controlled Exports to Russia and China

Russia and China both belong to BIS country Group D, but controlled exports to these two countries have followed very different trajectories. Figure C-1 shows the share of export value to Russia (panel a) and China (panel b) designated as EAR99 or given an explicit ECCN over the entire period 2009–2024. As with the broader Group D (Figure 1(c) in the main text), the share of export value subject to EAR99 has increased for both countries.

For exports to Russia, both EAR99 and ECCN shares exhibit notable volatility, particularly around 2014–2015 when sanctions were imposed following the annexation of Crimea. The dollar value of EAR99 exports to Russia actually fell dramatically—from about $1,314 million in 2010 to about $417 million by 2024—but, as a share of all exports to Russia, EAR99 exports increased steadily, from 22% to about 80%.

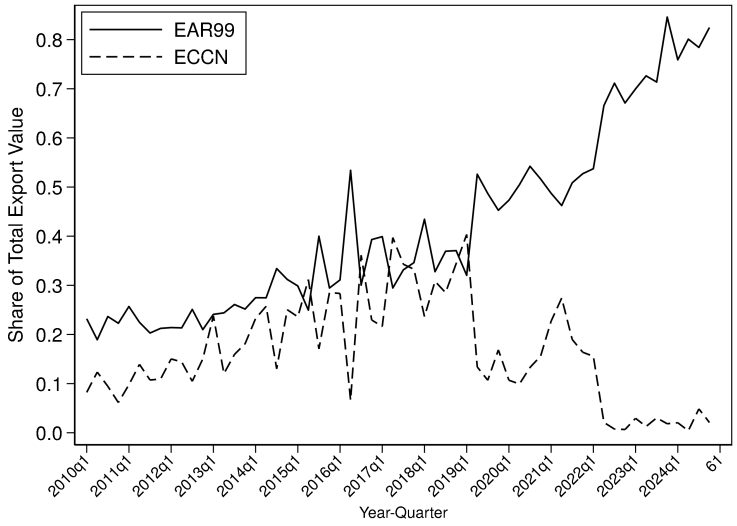
The share of export value to Russia accounted for by shipments with ECCNs—which peaked around 25% in 2013–2014—declined sharply to near zero by 2022, coinciding with the full-scale invasion of Ukraine and subsequent comprehensive sanctions. The export share of ECCNs exhibited a modest (and noisy) increase between 2010 and 2018, followed by a decrease. ECCN exports as a share of all exports to Russia did not drop meaningfully in 2014, when only eight ECCNs were included in the targeted list. However, there was a sharp drop in 2019 and again in 2022. The first decline followed the implementation, in 2018 and 2019, of a policy of presumption of denial for new applications for export licenses for goods that are controlled for their chemical and biological applications. The second decline followed a sweeping set of export controls added in 2022, including restrictions on products that did not previously require export licenses when destined for Russia (Congressional Research Service, 2024). The 2022 export-control measures imposed new license requirements on any goods classified under any ECCNs in categories 3 through 9 of the CCL under a general policy of denial (U.S. Department of Commerce, Bureau of Industry and Security, March 3, 2022). Consequently, the dollar value of ECCN exports to Russia fell to near zero.

The pattern for China shows a more gradual but sustained increase in the EAR99 share, from roughly 21% in 2010 to 50% in 2024. The ECCN share to China follows a similar upward trajectory through the mid-2010s, rising from about 13% to peaks near 30%, before declining to around 20% by 2024, likely reflecting the tightening of semiconductor and advanced technology export controls beginning in 2018–2019 and intensifying in 2022–2023.

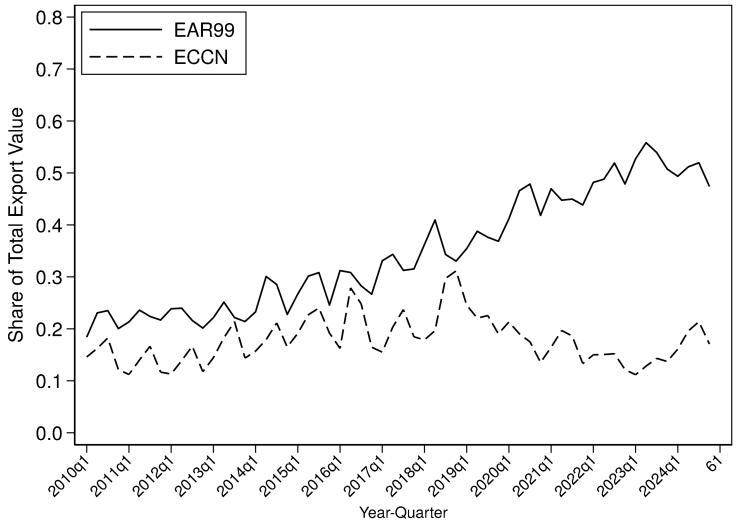
For China, the share of EAR99 exports started increasing noticeably after 2018. Since 2018, the U.S. government has increased restrictions on dual-use trade with China due to concerns about China's Military-Civil Fusion program (Congressional Research Service, 2022).[24]

[24]Military-Civil Fusion is the Chinese Communist Party's national strategy to build the country into a military superpower by fusing the country's military and civilian industrial resources (U.S. Department of State, 2021).

Figure C-1. Controlled Exports as Share of All U.S. Exports to Russia and China, 2010–2024



(a) Russia



(b) China

Notes: Value accounted for by shipments designated as EAR99, and shipments designated with all other ECCNs, as a share of all U.S. exports, by country group (defined in Section 2) and country. ECCN designates shipments that are subject to the EAR and are listed in the CCL. EAR99 designates shipments that are subject to the EAR but are not listed in the CCL.
Source: Authors' calculations using U.S. Census Bureau (2024) and country group designations from U.S. Department of Commerce, Bureau of Industry and Security (2025j).

D Identifying New Export Licenses in Census Data

The LFTTD includes the license type for all controlled shipments and license number for a subset of shipments with licenses.

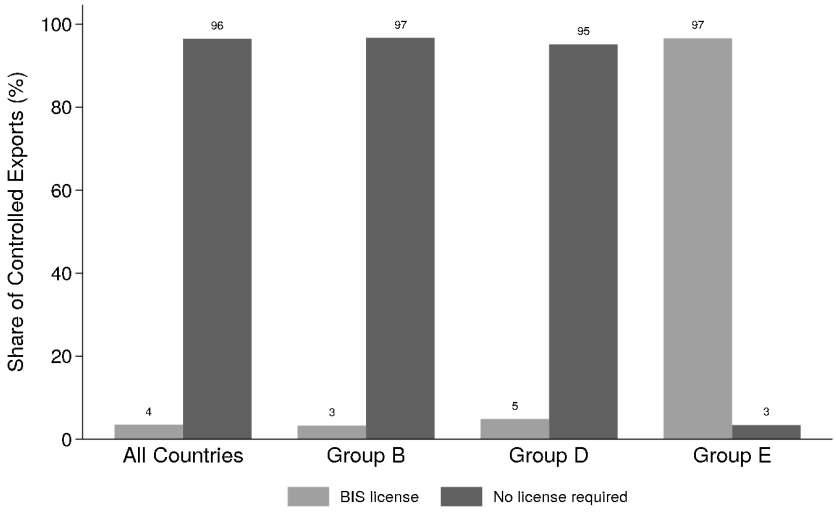
License type is a required field in the Electronic Export Information (EEI) that U.S. exporters are required to file (U.S. Census Bureau, 2025a; see U.S. Customs and Border Protection, 2025, for a list of all license types).[25] Figure D-1 plots the share of controlled exports by license requirement (BIS license vs. no license required) across country groups in 2024. Consistent with the prevalence of EAR99 designations documented in Section 2, 95% or more of controlled exports, by value, do not require explicit licenses across all country groups. The exception is Group E countries where the vast majority of exports do require a BIS license.

The license number is not a required field in the EEI and is therefore populated only in some cases. It could be populated for some shipments with a given license and not for others, and the license number may be incorrect (e.g., not follow the required pattern). We use published BIS licenses granted by year and compare the number to the number of licenses that appear in the LFTTD for the first time in a given year to provide a sense of how many licenses are missing in the LFTTD.

Figure D-2 plots the number of new licenses issued by the BIS alongside the first instances of BIS license usage observed in the LFTTD. The level difference between the aggregate totals and the microdata observations reflects the missing licenses in the LFTTD: in any given year, the LFTTD contains only approximately half the number of new licenses we expect to see based on BIS licenses granted. Even this is probably an over-estimate, because some of the licenses in the LFTTD may be phantom licenses: they may either be entry errors in shipments that do not have a license at all, or they may have a small error and appear to be new licenses when, in fact, the shipment was exported under a different license that was previously recorded.

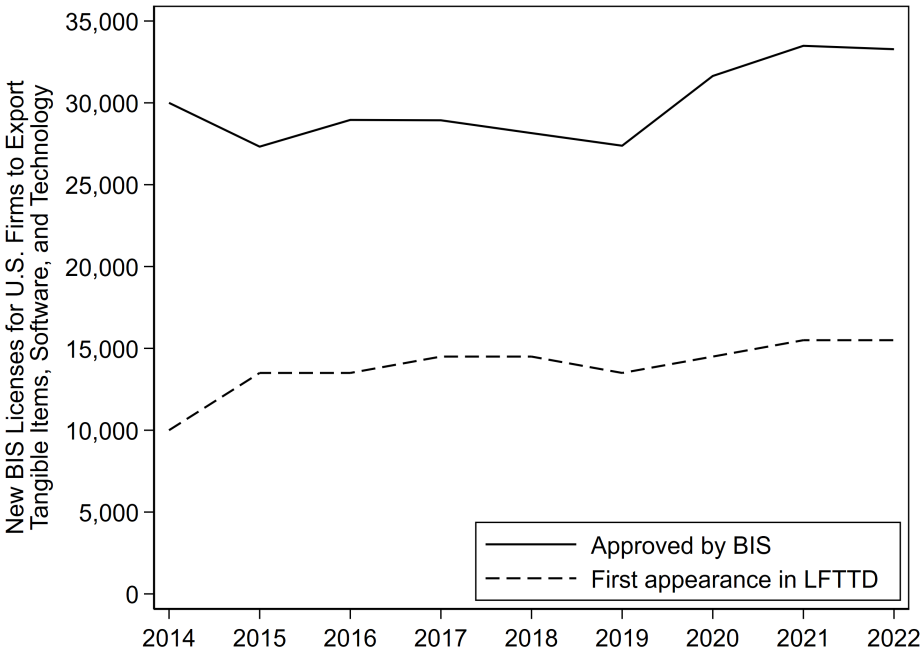
[25] A license type code beginning with a "C" indicates that it requires BIS review.

Figure D-1. License Type by Country Group, 2024



Notes: Value accounted for by shipments designated with a specific BIS license type (BIS license) or no license required, as a share of all controlled exports, by country group. Calculations exclude exports to Canada, Puerto Rico, and the U.S. Virgin Islands.
Source: Authors' calculations using U.S. Census Bureau (2024) and country group designations from U.S. Department of Commerce, Bureau of Industry and Security (2025j).

Figure D-2. New Licenses: BIS vs. LFTTD



Notes: We restrict LFTTD data to licenses issued by BIS (license type begins with letter C) for closest comparability. We omit license to Canada from BIS data because those are missing in LFTTD.
Source: Authors' calculations using LFTTD and U.S. Department of Commerce, Bureau of Industry and Security (2018, 2021).

E Russian Industry Sector Sanctions

E.1 Background

The U.S.'s 2014 Russia-targeted export controls restricted exports of both goods and services to Russia.[26] FR 45675 lists the products subject to controls under the Russian Industry Sector Sanctions. It states that "[t]he license review policy for all items requiring a license for export to Russia is presumption of denial when there is potential for use directly or indirectly for the exploration or production from deepwater (greater than 500 feet), Arctic offshore, or shale projects in Russia that have the potential to produce oil" (FR 45675, p. 45677). The FR notice also states that these items

> include, but are not limited to, drilling rigs, parts for horizontal drilling, drilling and completion equipment, subsea processing equipment, Arctic-capable marine equipment, wireline and down hole motors and equipment, drill pipe and casing, software for hydraulic fracturing, high pressure pumps, seismic acquisition equipment, remotely operated vehicles, compressors, expanders, valves, and risers.

Table E-1 summarizes the five main BIS rulemaking actions.

E.2 Targeted Goods Exports

We select all shipments of the ECCN (Table E-2) and Schedule B (Table E-3) codes specified in FR-45675 to Russia in the 2012 and 2013 to classify treated firms. In practice, ECCNs constitute a small fraction of these targeted exports. Approximately 98% of shipment value of targeted product exports to Russia in 2012 and 2013 was accounted for by the targeted Schedule B codes and either no ECCN or a non-targeted ECCN code; the remaining 2% is mostly accounted for by shipments with a targeted ECCN and a non-targeted Schedule B code (authors' calculations from U.S. Census Bureau, 2024). Only a very small sliver (less than 0.01% by value) were targeted both on Schedule B and on ECCNs. However, as indicated in Table E-2, only six of the eight targeted ECCN codes existed prior to 2014, so it is possible that some targeted exports to Russia were not tagged with ECCNs prior to 2014. Nearly half of all export shipments of targeted products to Russia pre-2014, by value, are accounted for by a single Schedule B code: 8431438010 (parts of oil and gas field machinery of subheading 8430.49, except parts of offshore drilling and production platforms). We select all shipments of these specified ECCNs or the Schedule B codes to China during the same period to classify control firms. Firms that exported targeted goods to both Russia and China in the pre-period are excluded from the control group.

[26] The Russia-targeted export controls also imposed restrictions, typically a denial of export licenses, on foreign individuals and businesses included in the "Entity List" maintained by BIS under Supplement No. 4 to part 744 of EAR (U.S. Department of Commerce, Bureau of Industry and Security, 2023). The foreign recipient of a shipment is not available in the LFTTD, so we cannot separately examine shipments to these foreign recipients.

E.3 Targeted Services Exports

To identify services exports potentially impacted by the 2014 Russia-targeted export controls, we use publicly available statistics on services export flows published by the U.S. Bureau of Economic Analysis. We distinguish targeted services from other services exports based on our reading of FR 45675. In addition to manufactured goods, the export controls covered software and data that pertain to the development, production, or use of the designated goods indicated as follows:[27]

> ECCN 0A998 is added to control specific oil and gas exploration items, including software and data; and
>
> ECCN 8D999 is added to the Commerce Control List to control software specially designed for the operation of unmanned vessels used in the oil and gas industry of Russia.

Guided by the notice, we categorize the following types of services as having been targeted:

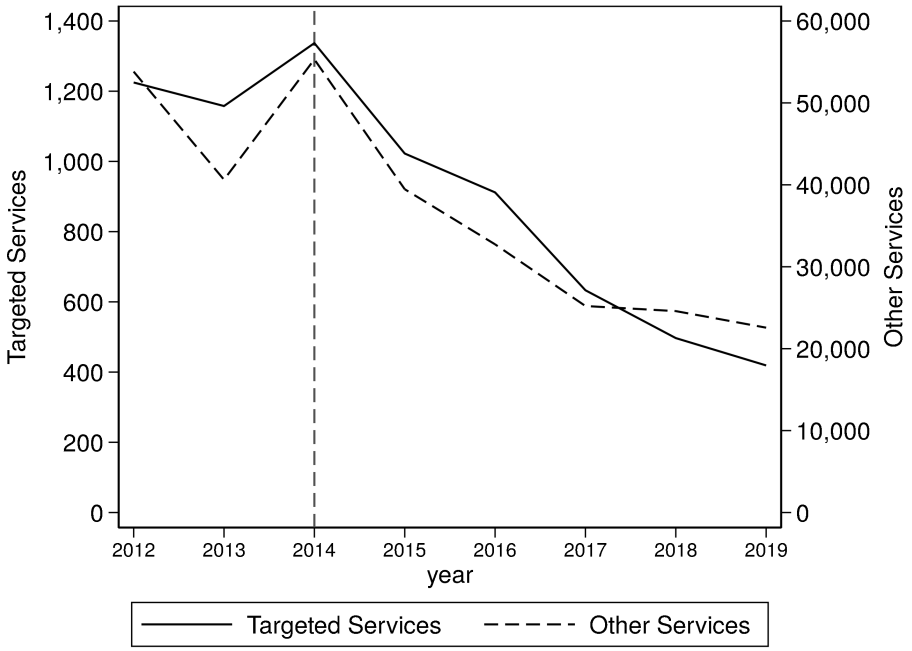
- Maintenance and repair services
- Construction abroad
- Licenses to reproduce and/or distribute computer software
- Computer software, including end-user licenses and customization
- Cloud computing and data storage services
- Database and other information services
- Engineering services
- Scientific and other technical services
- Services incidental to mining, and oil and gas extraction
- Operating leasing services.

Figure E-1 shows the export value of targeted and other services to Russia between 2012 and 2019. The dashed vertical line marks 2014. Before 2014, the value of targeted services exports to Russia is relatively flat. However, after the Russia-targeted export controls were implemented, targeted services exports to Russia declined markedly. The value of exports of other (non-targeted) services also declined, consistent with an overall chilling effect.[28] However, the value of other services exports flattens out after 2017, whereas targeted services exports exhibit a declining trend through 2019.

[27] See Kamal and Kroff (2026) for a discussion of ECCNs to identify goods that embody software, technology, and related intangible characteristics.

[28] The relatively high level of aggregation in services data likely causes some misclassification of targeted services into the non-targeted category.

Figure E-1. Targeted vs. Non-Targeted Services Exports to Russia, 2012-2019



Notes: Value of targeted and non-targeted services exports to Russia. Targeted services are defined in Appendix E.3; other services encompass all other services. Values are in millions of 2019 U.S. dollars.
Source: Authors' calculations using U.S. exports of services to Russia (U.S. Bureau of Economic Analysis, 2024) and FR 45675.

Table E-1. U.S. Bureau of Industry and Security Russian Sanctions Timeline, 2014

Action	Federal Register Notice	Effective Date
Addition of Person to the Entity List	79 FR 21394	April 16, 2014
Addition of Certain Persons to the Entity List	79 FR 24558	May 1, 2014
Addition of Certain Persons to the Entity List	79 FR 42452	July 22, 2014
Russian Oil Industry Sanctions and Addition of Person to the Entity List	79 FR 45675	August 6, 2014
Addition of Persons to the Entity List and Restrictions on Certain Military End Uses and Military End Users	79 FR 55608	September 17, 2014

Notes: Federal Register notices related to Russian sanctions under the jurisdiction of U.S. Bureau of Industry and Security.
Source: Authors' compilation from U.S. Department of Commerce, Bureau of Industry and Security (2014).

Table E-2. Targeted Export Control Classification Numbers, 2014

ECCN	Description
0A998[a]	Oil and gas exploration equipment, software, and data
3A229	Firing sets and equivalent high-current pulse generators for detonators controlled by 3A332
3A231	Neutron generator systems, including tubes
3A232	Detonators and multipoint initiation systems
6A991	Marine or terrestrial acoustic equipment
8A992	Vessels, marine systems or equipment not controlled by 8A001 or 8A002
8D992[a]	"Software" "specially designed" or modified for the "development," "production" or "use" of equipment controlled by 8A992

Notes: ECCNs included in the 2014 Russia-targeted export controls.
Source: FR 45675. ECCN descriptions obtained from U.S. Department of Commerce, Bureau of Industry and Security (2025h).
[a] ECCNs created in 2014 for the purposes of the Russia-targeted export controls. Quotations marks in original description.

Table E-3. Targeted Schedule B Codes, 2014

Schedule B	Description
7304110000	Line pipe of a kind used for oil or gas pipelines, seamless, of stainless steel
7304191020	Line pipe of a kind used for oil or gas pipelines, seamless, of iron (noncast) or nonalloy steel, with an outside diameter not exceeding 114.3 mm
7304191050	Line pipe for oil or gas pipelines, seamless, iron (noncast) or nonalloy steel, with outside diameter over 114.3 mm but not over 406.4 mm
7304191080	Line pipe of a kind used for oil or gas pipelines, seamless, of iron (noncast) or nonalloy steel, with an outside diameter exceeding 406.4 mm
7304195020	Line pipe of a kind used for oil or gas pipelines, seamless, of other alloy steel, not stainless, with an outside diameter not exceeding 114.3 mm
7304195050	Line pipe, used for oil or gas pipelines, seamless, of other alloy steel, not stainless, with an outside diameter >114.3 mm, but <406.4 mm
7304195080	Line pipe of a kind used for oil or gas pipelines, seamless, of alloy steel, not stainless, with an outside diameter exceeding 406.4 mm
7304220000	Oil well drill pipe, of stainless steel
7304233000	Oil well drill pipe, of iron or nonalloy steel
7304236000	Oil well drill pipe, of alloy steel other than stainless steel
7304241000	Oil well casing of stainless steel
7304246000	Oil well tubing of stainless steel
7304291055	Oil well casing of iron or nonalloy steel
7304293155	Oil well casing of other alloy steel not stainless
7304295000	Oil well tubing of iron or nonalloy steel
7304296100	Oil well tubing of other alloy steel other than stainless steel
7305111000	Line pipe for oil or gas, longitudinally submerged arc welded, external diameter more than 406.4 mm, circular cross-sections, of iron or nonalloy steel
7305115000	Line pipe for oil/gas pipelines, longitudinally submerged arc welded with external diameter over 406.4 mm, of alloy steel, with circular cross-section
7305121000	Line pipe for oil or gas, other longitudinally welded, external diameter more than 406.4 mm, circular cross-section, iron or nonalloy steel
7305125000	Line pipe for oil or gas pipelines, longitudinally welded with external diameter >406.4 mm, of alloy steel, with circular cross section
7305191000	Line pipe for oil or gas other than longitudinally welded, external diameter more than 406.4 mm, circular cross-section, iron or nonalloy steel
7305195000	Line pipe for oil or gas pipelines, with external diameter >406.4 mm, of alloy steel, circular cross section, welded/riveted, NESOI
7305203000	Casing, oil or gas drilling, other than seamless, circular cross-section, external diameter over 406.4 mm, iron or nonalloy steel
7305207000	Casing, oil or gas drilling, other than seamless, circular cross-section, external diameter over 406.4 mm, alloy steel
7306110000	Line pipe for oil or gas not seamless NESOI, of stainless steel
7306191000	Line pipe for oil or gas not seamless NESOI, of iron or nonalloy steel

Notes: List of Schedule B codes subject to the 2014 Russia-targeted export controls.
Source: FR 45675.

Schedule B	Description
7306195000	Line pipe for oil or gas not seamless NESOI, of alloy steel other than stainless steel
7311000000	Containers for compressed or liquefied gas of iron or steel
7613000000	Aluminum containers for compressed or liquefid gas
8207130000	Rock drilling or earth boring tools with working part of cermets, and parts thereof
8207191030	Percussion rock drill bits, core bits and reamers, of base metal, and parts thereof
8207192030	Rotary rock drill bits, core bits and reamers of base metal, and parts thereof
8207195030	Rock drilling or earth boring tools of base metals, NESOI, and parts thereof
8413500010	Oil well and oil field pumps, reciprocating positive displacement
8413600050	Oil well and oil field pumps, rotary positive displacement
8413820000	Liquid elevators
8413920000	Parts of liquid elevators
8421398020	Electrostatic precipitators, industrial gas cleaning equipment
8421398030	Industrial gas cleaning equipment, NESOI
8421398040	Gas separation equipment
8430494000	Offshore oil and natural gas drilling and production platforms
8430498010	Boring or sinking machinery, rotary, for oil well and gas field drilling
8430498020	Boring or sinking machinery for oil well and gas field drilling, NESOI
8431390050	Parts suitable for use solely or principally with the oil and gas field machinery of headings 8425 to 8430
8431434000	Offshore oil and natural gas drilling and production platform parts, of subheading 8430.41 or 8430.49
8431438010	Parts of oil and gas field machinery of subheading 8430.49 except parts of offshore drilling and production platforms
8431438090	Parts of boring or sinking machinery of subheading 8430.41 or 8430.49, NESOI
8479899850	Oil and gas field wire line and downhole equipment
8705200000	Mobile drilling derricks
8708998175	Parts and accessories, for motor vehicles of heading 8705.20, NESOI
8905200000	Floating or submersible drilling or production platforms
8905901000	Floating docks

Notes: List of Schedule B codes subject to the 2014 Russia-targeted export controls.

Source: FR 45675.

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